Drawing Analysis Results

AI Analysis Summary

Plan View: R3.0 (page 3)

Scale: 1:70

roof_drains: 4

Detail refs: Detail 3/R4.0, Detail 2/R4.0, Detail 4/R4.0, Detail 5/R4.0, Detail 2/R3.0

Details: R2.0 (page 2)

> **DEMOLITION PLAN** [slope_plan] - measured in: sqft
>
> - 1. EXISTING ASPHALT ROOFING MEMBRANE → SBS_Membrane (Remove from within planters)
> - 2. CONCRETE TOPPING → None (Remove and dispose of entire concrete topping, typ.)
> - 3. PLANTER WALLS AND WATERPROOFING → Base_Membrane (Remove and dispose)
> - 4. EXISTING FLASHINGS → Flashing_General (Remove and dispose)

> **PLANTER WALL DEMO DETAIL** [parapet] - measured in: linear_ft
>
> - 1. STRUCTURAL CONCRETE SLAB → None (Existing, to remain)
> - 2. RUBBERIZED ASPHALTIC WATERPROOF MEMBRANE → Base_Membrane (Part of typical existing roof assembly to be removed)
> - 3. POLYETHYLENE SEPARATION SHEET → Vapour_Barrier_Membrane (Part of typical existing roof assembly to be removed)
> - 4. ±38mm XPS INSULATION → XPS_Insulation (Part of typical existing roof assembly to be removed)
> - 5. ±100mm CONCRETE TOPPING → None (Part of typical existing roof assembly to be removed)
> - 6. EXISTING PLANTER WALLS → None (700mm height shown. Remove and dispose.)
> - 7. EXISTING PLANTER WATERPROOFING → Base_Membrane (Remove and dispose)
> - 8. EXISTING FLASHINGS → Flashing_General (Remove and dispose)

Details: R4.0 (page 4)

> **TYPICAL WALL UPTURN DETAIL** [parapet] - measured in: linear_ft
>
> - 1. 26GA GALVANIZED STEEL BREAKSHAPE → Flashing_General (To facilitate membrane transition)
> - 2. 6.4mm ø PIN BOLTS @ 300mm O/C → Fasteners (Fasten brakeshape into pre-cast)
> - 3. 25mm CONT. CLAMPING BAR → Clips (Terminate membrane at upturn)

- 4. 38mm HEX HEAD SCREWS @ 400mm O/C → Screws (Fasten clamping bar)
- 5. 24GA PRE-FINISHED STEEL FLASHING → Flashing_General (To cover termination bar)
- 6. SEALANT → Sealant_General (Fill 12mm deep sawcut reglet and top lip)
- 7. SEALANT AT GLAZING TO PRECAST TRANSITION → Sealant_General (Remove and replace existing)

TYPICAL PARAPET UPTURN DETAIL [parapet] - measured in: linear_ft

- 1. MINERAL WOOL INSULATION → Batt_Insulation (Fill void)
- 2. 13mm PLYWOOD SHEATHING → Plywood_Sheathing (Screwed at top and bottom @ 300mm O/C over 18ga steel stud curb)
- 3. SELF-ADHERED AIR/VAPOUR BARRIER MEMBRANE → Vapour_Barrier_Membrane (Strip at inside face of curb)
- 4. SELF-ADHERED SANDED STRIPPING MEMBRANE → Base_Membrane_Peel_Stick (Applied horizontally to minimize joints, leave 25mm loop)
- 5. SELF-ADHERED CAP SHEET MEMBRANE → Cap_Membrane (Cover stripping membrane)
- 6. 25mm CONT. CLAMPING BAR → Clips (Terminate membrane at upturn)
- 7. 38mm HEX HEAD SCREWS @ 400mm O/C → Screws (Fasten clamping bar)
- 8. 24GA PRE-FINISHED STEEL FLASHING → Flashing_General (To cover curb)
- 9. 24GA PRE-FINISHED STEEL FLASHING → Flashing_General (To cover termination bar)
- 10. SEALANT → Sealant_General (Finish at existing reglet)

EXISTING ROOF DRAIN WATERPROOFING [drain] - measured in: each

- 1. WOOD BLOCKING → Wood_Blocking_Lumber (At drain perimeter for flange support)
- 2. MINERAL WOOL INSULATION → Batt_Insulation (Between new drain body and concrete slab)
- 3. NEW DRAIN BODY AND PIPE → Roof_Drain (Tie into existing roof drainage system)
- 4. VAPOUR BARRIER → Vapour_Barrier_Membrane (Back wrap)
- 5. FLEECE REINFORCED PMMA MEMBRANE → Fleece_Reinforcement_Fabric (Within sump and 200mm past on all sides)
- 6. PERIMETER SEALANT → Sealant_General (Compatible with roof membrane)
- 7. CLAMPING RING → Roof_Drain (Secure membrane)
- 8. DRAIN DOME → Roof_Drain (Top of drain assembly)

TYPICAL PRECAST CONCRETE PANEL JOINT SEALANT REPLACEMENT [expansion_joint] - measured in: linear_ft

- 1. CLOSED CELL BACKING ROD → EPDM_Accessory (Backing for sealant)
- 2. SEALANT → Sealant_General (Joint width minimum 12mm)

DOOR THRESHOLD DETAIL [parapet] - measured in: linear_ft

- 1. FLEECE REINFORCED LIQUID APPLIED MEMBRANE → Fleece_Reinforcement_Fabric (Below door threshold)
- 2. ADDITIONAL LAYER OF CAP SHEET MEMBRANE → Cap_Membrane (At door threshold)
- 3. COMPRESSIBLE NEOPRENE GASKET → Sealant_General (Below door threshold, continuous along full door length)
- 4. DOOR THRESHOLD → Flashing_General (Remove and replace existing as required)

+15 WALL UPTURN DETAIL [parapet] - measured in: linear_ft

- 1. GRANULAR BENTONITE → Sealant_General (Pack void where 50mm of existing insulation removed)
- 2. NEW VAPOUR BARRIER MEMBRANE → Vapour_Barrier_Membrane (Lap min. 200mm onto existing membrane)
- 3. 13mm GLASS FACED GYPSUM SHEATHING → Gypsum_Fiber_Coverboard (Fasten into existing concrete topping)
- 4. VAPOUR BARRIER MEMBRANE → Vapour_Barrier_Membrane (Continue min. 50mm above insulation)
- 5. METAL CLADDING → Metal_Panel (Temporarily remove and reinstate)

Specification PDF Analysis (Deterministic Regex Engine)

20 unique products found across **7** categories — **10** mapped to pricing keys.
Project: Replacement

- ▶ **Membranes (5)**
- ▶ **Sealants & Adhesives (7)**
- ▶ **Drainage (3)**
- ▶ **Wood & Sheathing (1)**
- ▶ **Miscellaneous (1)**
- ▶ **Insulation (1)**
- ▶ **Metal Flashings & Accessories (2)**

Spec-Driven Pricing Estimate (Drawing spatial data + Specification materials — deterministic)

LINE ITEMS RESOLVED	RESOLUTION FAILURES	TOTAL MATERIAL COST
3	**0**	**$1,682.68**

Resolved Line Items

DETAIL	TYPE	CONFIRMED MATERIAL	QTY	UNIT PRICE	LINE COST	SOURCE
PLANTER WALL DEMO DETAIL R2.0	parapet	Vapour Barrier Membrane `Vapour_Barrier_Membrane` spec pg 68	1 roll	$173.66	$173.66	FAL
TYPICAL WALL UPTURN DETAIL R4.0	parapet	Sealant (General) `Sealant_General` spec pg 2, 86, 88, 89, 90, 91, 92, 93, 94, 95	7 tube	$29.86	$209.02	FAL
EXISTING ROOF DRAIN WATERPROOFING R4.0	drain	Roof Drain `Roof_Drain` spec pg 2, 13, 29, 56, 63, 73, 82, 96, 97, 98, 99	4 EA	$325.00	$1,300.00	FAL
				Total Material Cost	**$1,682.68**	

Detail-Based Quantity Takeoff (AI-Analyzed)

ROOF AREA	PERIMETER	PARAPET	PENETRATIONS
1,743 sqft	**238 LF**	**123 LF x 1.0 ft**	**0**

Field Assembly (Plan View) [field_assembly] (ref: ?)

Measured in: sqft | Base value: 1,743.0 | Detail cost: $28,462.90 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
Asphaltic Primer	8 pail	$282.83	$2,262.64
SBS Base Sheet - Field (Sopraply Base 520)	21 roll	$127.25	$2,672.25
SBS Base Sheet - Wall (Sopraply Base 520)	2 roll	$127.25	$254.50
SBS Cap Sheet - Field (Sopraply Traffic Cap)	24 roll	$150.00	$3,600.00

MATERIAL	QTY	UNIT PRICE	LINE COST
SBS Cap Sheet - Wall (Sopraply Traffic Cap)	2 roll	$150.00	$300.00
Tapered ISO Insulation (drainage slope)	1918 sqft	$3.10	$5,945.80
XPS Insulation (Sopra-XPS 40 Type 4)	120 sheet (2'x8')	$43.70	$5,244.00
Drainage Board (Sopradrain EcoVent)	7 roll (6'x50')	$410.53	$2,873.71
Filter Fabric	7 roll	$150.00	$1,050.00
Soprasmart ISO HD 1/2" (factory laminated coverboard)	80 sheet (3'x8')	$53.25	$4,260.00

PLANTER WALL DEMO DETAIL [parapet] (ref: R2.0)

Measured in: linear_ft | Base value: 123.0 | Detail cost: $1,223.66 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
STRUCTURAL CONCRETE SLAB	136 unit	$0.00	-
RUBBERIZED ASPHALTIC WATERPROOF MEMBRANE	0 roll	$0.00	-
POLYETHYLENE SEPARATION SHEET	1 roll	$173.66	$173.66
±38mm XPS INSULATION	0 sheet (2'x8')	$0.00	-
±100mm CONCRETE TOPPING	0 unit	$0.00	-
EXISTING PLANTER WALLS	0 unit	$0.00	-
EXISTING PLANTER WATERPROOFING	0 roll	$0.00	-
EXISTING FLASHINGS	14 10ft piece	$75.00	$1,050.00

TYPICAL WALL UPTURN DETAIL [parapet] (ref: R4.0)

Measured in: linear_ft | Base value: 123.0 | Detail cost: $988.94 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
26GA GALVANIZED STEEL BREAKSHAPE	0 10ft piece	$0.00	-
6.4mm ø PIN BOLTS @ 300mm O/C	2 box (1M)	$350.00	$700.00
25mm CONT. CLAMPING BAR	0 piece	$10.19	-
38mm HEX HEAD SCREWS @ 400mm O/C	2 box	$39.96	$79.92
24GA PRE-FINISHED STEEL FLASHING	0 10ft piece	$0.00	-
SEALANT	7 tube	$29.86	$209.02
SEALANT AT GLAZING TO PRECAST TRANSITION	0 tube	$0.00	-

TYPICAL PARAPET UPTURN DETAIL [parapet] (ref: R4.0)

Measured in: linear_ft | Base value: 123.0 | Detail cost: $909.66 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
MINERAL WOOL INSULATION	4 bundle	$83.99	$335.96
13mm PLYWOOD SHEATHING	5 4'x8' sheet	$48.74	$243.70
SELF-ADHERED AIR/VAPOUR BARRIER MEMBRANE	0 roll	$0.00	-
SELF-ADHERED SANDED STRIPPING MEMBRANE	2 roll	$165.00	$330.00
SELF-ADHERED CAP SHEET MEMBRANE	0 roll	$0.00	-
25mm CONT. CLAMPING BAR	0 piece	$0.00	-
38mm HEX HEAD SCREWS @ 400mm O/C	0 box	$0.00	-
24GA PRE-FINISHED STEEL FLASHING	0 10ft piece	$0.00	-
24GA PRE-FINISHED STEEL FLASHING	0 10ft piece	$0.00	-

MATERIAL	QTY	UNIT PRICE	LINE COST
SEALANT	0 tube	$0.00	-

EXISTING ROOF DRAIN WATERPROOFING [drain] (ref: R4.0)

Measured in: each | Base value: 4 | Detail cost: $1,316.59 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
WOOD BLOCKING	1 8ft piece	$16.59	$16.59
MINERAL WOOL INSULATION	0 bundle	$0.00	-
NEW DRAIN BODY AND PIPE	4 EA	$325.00	$1,300.00
VAPOUR BARRIER	0 roll	$0.00	-
FLEECE REINFORCED PMMA MEMBRANE	0 roll	$0.00	-
PERIMETER SEALANT	0 tube	$0.00	-
CLAMPING RING	0 EA	$0.00	-
DRAIN DOME	0 EA	$0.00	-

TYPICAL PRECAST CONCRETE PANEL JOINT SEALANT REPLACEMENT [expansion_joint] (ref: R4.0)

Measured in: linear_ft | Base value: 59.5 | Detail cost: $398.99 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
CLOSED CELL BACKING ROD	1 roll	$398.99	$398.99
SEALANT	0 tube	$0.00	-

DOOR THRESHOLD DETAIL [parapet] (ref: R4.0)

Measured in: linear_ft | Base value: 123.0 | Detail cost: $0.00 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
FLEECE REINFORCED LIQUID APPLIED MEMBRANE	0 roll	$0.00	-
ADDITIONAL LAYER OF CAP SHEET MEMBRANE	0 roll	$0.00	-
COMPRESSIBLE NEOPRENE GASKET	0 tube	$0.00	-
DOOR THRESHOLD	0 10ft piece	$0.00	-

+15 WALL UPTURN DETAIL [parapet] (ref: R4.0)

Measured in: linear_ft | Base value: 123.0 | Detail cost: $1,637.30 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
GRANULAR BENTONITE	0 tube	$0.00	-
NEW VAPOUR BARRIER MEMBRANE	0 roll	$0.00	-
13mm GLASS FACED GYPSUM SHEATHING	5 sheet (4'x8')	$29.62	$148.10
VAPOUR BARRIER MEMBRANE	0 roll	$0.00	-
METAL CLADDING	136 lin ft	$10.95	$1,489.20

Total Material Cost **$34,938.04**

Standard Estimate (for comparison)

Standard Total Estimate	**$63,261.06**